COCONTINUITY OF THE FUNCTOR U |-> U_PH FROM AFFINES OVER S INTO THE BIG PH SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2913, 2921] and proof(s) [[2923, 2939]]; this spec covers the complete environment instead (statement 2913-2921, proof span(s) [(2923, 2939)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original reference 03CG: sources/stacks-chow/tags.txt, source lines 4383–4383. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03A0: sources/stacks-chow/tags.txt, source lines 4291–4291. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let Sch_{ph} be a big ph site containing S. The functor $(Aff/S)_{ph} \to (Sch/S)_{ph}$ is cocontinuous and induces an equivalence of topoi from $Sh((Aff/S)_{ph})$ to $Sh((Sch/S)_{ph})$.*

Proof. The notion of a special cocontinuous functor is introduced in Sites, Definition 03CG. Thus we have to verify assumptions (1) – (5) of Sites, Lemma 03A0. Denote the inclusion functor $u : (Aff/S)_{ph} \to (Sch/S)_{ph}$. Being cocontinuous follows because any ph covering of T/S, T affine, can be refined by a standard ph covering of T by definition. Hence (1) holds. We see u is continuous simply because a finite ph covering of an affine by affines is a ph covering. Hence (2) holds. Parts (3) and (4) follow immediately from the fact that u is fully faithful. And finally condition (5) follows from the fact that every scheme has an affine open covering (which is a ph covering). □

Date: stacks-project:topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.